

Week 9 Tutorial

Introduction

This workshop will help you familiarize yourself with autoregressive conditional heteroskedasticity (ARCH) models, as well as testing for the adequacy of these models.

Work in your groups to come up with strategies and answers for the blank sections. You have all of the code, we need you to interpret it. This helps for your assignment.

Question 1

Using the data in [IBM_daily_2024.xlsx](#) and [IBM_monthly_2024.xlsx](#), generate daily and monthly returns on IBM using the corresponding closing stock price. Use these data to answer the following questions.

1. Estimate an ARMA(1,1) model for both daily and monthly returns. Does the ARMA(1,1) model appear to fit the daily or monthly return series better? *HINT: use the residual correlograms.*

With time series data, we do not know what a sensible lag structure is, so we usually fit a few models. What we hope is that we capture all of the time-dynamics in our model, which means that there should be no serial correlation in the residuals. This can be an iterative process.

Let's assume that we estimated the constant mean equation for both series of returns and found evidence of persistence in the residuals. We then fit some AR models, however we could not remove all of the persistence without using a high order for p.

The next step would be to try using an ARMA(1,1) model. This is an extension of the AR(p) model. Generally, if we find evidence of a high-order AR model in returns, an ARMA model can be used to capture the time dynamics more efficiently.

The ARMA(1,1) model is defined as:

$$r_t = c + \rho r_{t-1} + \epsilon_t + \theta \epsilon_{t-1}$$

The ARMA(1,1) model for the daily data and monthly data is:

```
library(xts)

# Daily data

d.df <- readxl::read_excel("ibm_daily_2024.xlsx")

# daily close convert to xts object
d.close.xts = as.xts(d.df)

# log return - in percentage
lretd <- na.omit(diff(log(d.close.xts)))*100
colnames(lretd) <- "lretd"
# head(lretd)
# tail(lretd)

# ARMA(1,1)

fit_armad <- arima(as.numeric(lretd), order = c(1,0,1),
                  include.mean = TRUE,
                  method = "CSS")

fit_armad
```

Call:

```
arima(x = as.numeric(lretd), order = c(1, 0, 1), include.mean = TRUE, method = "CSS")
```

Coefficients:

	ar1	ma1	intercept
	-0.3516	0.3139	0.0086
s.e.	0.2262	0.2320	0.0203

sigma² estimated as 2.671: part log likelihood = -11675.49

```
# residuals from ARMA(1,1)
res_d <- residuals(fit_armad)
```

```
# Monthly data
```

```

m.df <- readxl::read_excel("ibm_monthly_2024.xlsx")

# from Jan 1980 to May 2024
library(xts)
idx <- seq(from = as.Date("1980-01-01"),
           by   = "month",
           length.out = nrow(m.df))

close.xts <- xts( m.df, order.by = idx)

# convert "1980-01-01" to "Jan 1980"
index(close.xts) <- as.yearmon(index(close.xts))
# head(close.xts)
# tail(close.xts)

# log return - NOT in percentage
lretm <- na.omit(diff(log(close.xts)))
colnames(lretm) <- "lretm"
# head(lretm)
# tail(lretm)

# ARMA(1,1)

fit_armam <- arima(as.numeric(lretm), order = c(1,0,1),
                  include.mean = TRUE,
                  method = "CSS")

fit_armam

```

Call:

```
arima(x = as.numeric(lretm), order = c(1, 0, 1), include.mean = TRUE, method = "CSS")
```

Coefficients:

	ar1	ma1	intercept
	0.5211	-0.5495	0.0049
s.e.	0.2590	0.2556	0.0029

sigma^2 estimated as 0.005185: part log likelihood = 644.83

```

# residuals from ARMA(1,1)
res_m <- residuals(fit_armam)

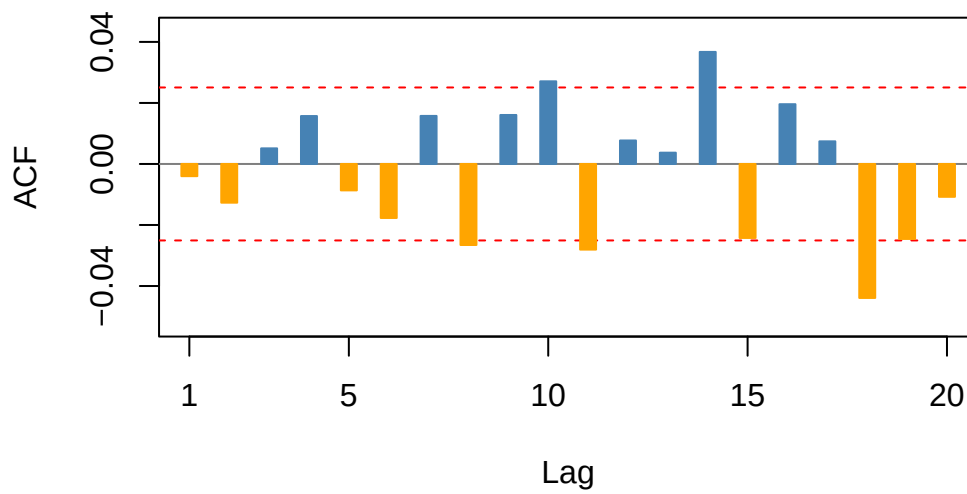
```

With time series data, we do not know what a sensible lag structure is, so we usually fit a few models. What we hope is that we capture all of the time-dynamics in our model, which means that there should be no serial correlation in the residuals.

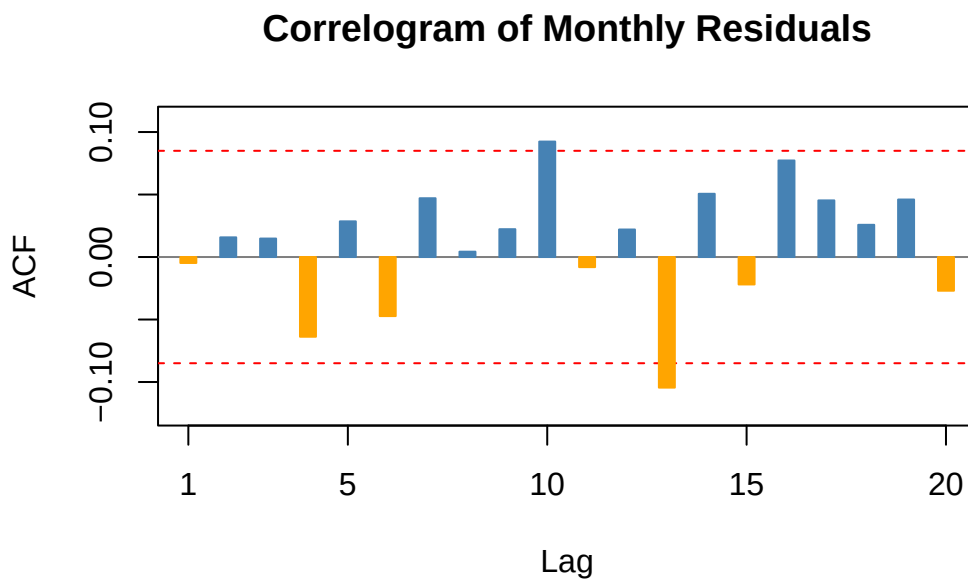
We could perform a serial-correlation LM test to see if we have adequately captured the time dynamics, and/or use the correlograms (the choice is yours). We keep estimating different models until we see no persistence. Here, we will use the correlograms below to check.

Lag	AC	Q	Prob
1	-0.0040	0.0979	7.544e-01
2	-0.0127	1.0824	5.821e-01
3	0.0051	1.2426	7.428e-01
4	0.0157	2.7509	6.003e-01
5	-0.0087	3.2090	6.678e-01
6	-0.0177	5.1259	5.278e-01
7	0.0157	6.6434	4.669e-01
8	-0.0266	10.9603	2.040e-01
9	0.0161	12.5383	1.846e-01
10	0.0271	17.0333	7.363e-02
11	-0.0281	21.8585	2.549e-02
12	0.0077	22.2216	3.511e-02
13	0.0037	22.3052	5.081e-02
14	0.0367	30.5710	6.363e-03
15	-0.0242	34.1744	3.218e-03
16	0.0196	36.5266	2.444e-03
17	0.0074	36.8637	3.511e-03
18	-0.0439	48.6875	1.189e-04
19	-0.0246	52.3893	5.784e-05
20	-0.0108	53.1005	7.855e-05

Correlogram of Daily Residuals



Lag	AC	Q	Prob
1	-0.0048	0.0121	0.9125
2	0.0157	0.1446	0.9303
3	0.0148	0.2616	0.9671
4	-0.0638	2.4498	0.6536
5	0.0285	2.8887	0.7171
6	-0.0473	4.0946	0.6639
7	0.0470	5.2912	0.6245
8	0.0043	5.3010	0.7250
9	0.0223	5.5702	0.7820
10	0.0924	10.2127	0.4220
11	-0.0081	10.2485	0.5082
12	0.0220	10.5128	0.5711
13	-0.1045	16.4888	0.2237
14	0.0506	17.8921	0.2117
15	-0.0219	18.1564	0.2545
16	0.0772	21.4389	0.1622
17	0.0453	22.5705	0.1638
18	0.0258	22.9372	0.1930
19	0.0460	24.1093	0.1920
20	-0.0270	24.5124	0.2207



what do these correlograms tell us?

2. Conduct an ARCH(5) test for the model residuals of the ARMA(1,1) in **levels**. Carry out this test for both daily and monthly returns. What do the results of this imply?

Refer to the ARCH test handout for the appropriate steps. We have done this test in Week 6. This is a test for ARCH, not leftover ARCH. A test for left-over ARCH is the second test in the ARCH test handout.

In R we use the ArchTest function from the FinTS package to do this test. This will give us the p-value of the test only - you must remember to write out the steps if required.

```
# ARCH(5) test on residuals
# Daily
FinTS::ArchTest(res_d, lags = 5)
```

ARCH LM-test; Null hypothesis: no ARCH effects

```
data: res_d
Chi-squared = 434.35, df = 5, p-value < 2.2e-16
```

```
# Monthly
FinTS::ArchTest(res_m, lags = 5)
```

ARCH LM-test; Null hypothesis: no ARCH effects

data: res_m

Chi-squared = 17.503, df = 5, p-value = 0.003638

What do the test results suggest? What should you do next?

3. For both daily and monthly returns, fit an ARMA(1,1) model with ARCH(2) errors. That is, fit the model:

$$\begin{aligned}r_t &= c + \rho r_{t-1} + \epsilon_t + \theta \epsilon_{t-1}, \\ \epsilon_t &\sim N(0, \sigma_t^2), \\ \sigma_t^2 &= a_0 + a_1 \epsilon_{t-1}^2 + a_2 \epsilon_{t-2}^2.\end{aligned}$$

When we estimate ARCH models, it is important to remember that there are two stages. First we estimate log-returns in levels, which in this case is our ARMA(1,1) model. Then we define our ARCH model. We need to report all three lines of equations that were given above.

NOTE: The specification for the variance is deterministic (it has no error term) so assessing fit, using R^2 does not make sense in ARCH models. So we do not need to look at R^2 or the log-likelihood.

When we estimate the equation, we need to use the **rugarch** package. This requires us to specify the ARCH model, the mean equation and to define the distribution to be used.

You only need to change the orders where required. You could try different distributions for your project if you wish, as long as you can justify your choice.

```
# Specify the ARMA(1,1)-ARCH(2) model
library(rugarch)

spec_arma11arch2 <- ugarchspec(
  variance.model = list(
    model = "sGARCH",          # standard GARCH
    garchOrder = c(2, 0)      # ARCH(2)
  ),
  mean.model = list(
    armaOrder = c(1, 1),      # ARMA(1,1)
    include.mean = TRUE
  ),
  distribution.model = "norm" # normal errors
)
```

```
# Daily
fit_arma1larch2d <- ugarchfit(
  spec = spec_arma1larch2,
  data = lretd
)

# Monthly
fit_arma1larch2m <- ugarchfit(
  spec = spec_arma1larch2,
  data = lretm
)
```

In the specification code, we first specify our variance (or ARCH) in the **variance.model** section, then specify our level equation in the **mean.model** section (see the comments in the code). You can also select an error distribution if you wish. We will use the Normal distribution, but you could also use the t-distribution for example (you could try this in your Part B assignment).

The estimation results are below. To report these equations, substitute the estimate coefficients and their standard errors into the three equations defined above (as you normally do when reporting regressions).

NOTE: the omega in the output is a_0 in our notation.

The estimation output for the Daily series is:

```
# inspect results
fit_arma1larch2d@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.01563071	0.01683934	0.9282259	3.532904e-01
ar1	-0.89415978	0.14530408	-6.1537141	7.568908e-10
ma1	0.88631995	0.14875015	5.9584475	2.546455e-09
omega	1.34405382	0.04115181	32.6608716	0.000000e+00
alpha1	0.29309584	0.02371641	12.3583544	0.000000e+00
alpha2	0.27733948	0.02573755	10.7756768	0.000000e+00

The estimation output for the Monthly series is:

```
fit_arma1larch2m@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.005802506	0.002969145	1.9542682	0.05066950
ar1	-0.333618177	1.780071117	-0.1874185	0.85133255
ma1	0.317171761	1.846395467	0.1717789	0.86361136


```
omega    0.004012516 0.000369477 10.8599891 0.00000000
alpha1   0.120900185 0.053716891  2.2506922 0.02440504
alpha2   0.104027625 0.054140634  1.9214334 0.05467709
```

4. For both return series, conduct an ARCH(5) test to determine if any excess autocorrelation exists in the variances of the series at the 5% level.

*HINT: this is a test for **left-over ARCH** and does not have the same auxiliary regression as the test in part 2 - it uses standardised residuals.*

Since we have estimated ARCH(2), it does not make sense to test for leftover ARCH of order 2 or below. We want to know if our ARCH models have captured all of the ARCH effects. Remember that we do not know what a sensible lag structure is, so we let the data tell us.

Refer to the ARCH test handout for the required steps. Be sure to use the ARCH model equations. Since the test for left-over ARCH uses standardised residuals, we need to specify this in our code. If you compare this code to the ARCH test code from step earlier, you should notice a difference. The ARCH test for the daily series is:

```
# Q4 - left-over arch(5) test daily
# standardised residuals
res_s.arch2d    <- residuals(fit_arma11arch2d, standardize = TRUE)

# Left over ARCH(5) test
FinTS::ArchTest(res_s.arch2d, lags = 5)
```

ARCH LM-test; Null hypothesis: no ARCH effects

```
data:  res_s.arch2d
Chi-squared = 32.179, df = 5, p-value = 5.475e-06
```

For the Monthly series, we get:

```
# standardised residuals
res_s.arch2m    <- residuals(fit_arma11arch2m, standardize = TRUE)

# Left over ARCH(5) test
FinTS::ArchTest(res_s.arch2m, lags = 5)
```

ARCH LM-test; Null hypothesis: no ARCH effects

```
data:  res_s.arch2m
Chi-squared = 8.3902, df = 5, p-value = 0.136
```

What do the test results suggest? What should you do next?

5. For the **daily** returns, re-estimate the model but with ARCH(5) errors.

We don't know how many ARCH lags we need, so let's estimate an ARCH(5) model on the daily returns. The estimated output is below:

```
# re-estimate with ARCH(5) now

spec_arma1larch5d <- ugarchspec(
  variance.model = list(
    model = "sGARCH",          # standard GARCH
    garchOrder = c(5, 0)      # ARCH(5)
  ),
  mean.model = list(
    armaOrder = c(1, 1),      # ARMA(1,1)
    include.mean = TRUE
  ),
  distribution.model = "norm" # normal errors
)

# 2. Estimate the ARMA-GARCH models
fit_arma1larch5d <- ugarchfit(
  spec = spec_arma1larch5d,
  data = lretd
)

# inspect results

fit_arma1larch5d@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.03062382	0.015714878	1.948715	5.132942e-02
ar1	-0.94951637	0.004003339	-237.181112	0.000000e+00
ma1	0.94864734	0.005757386	164.770506	0.000000e+00
omega	0.87623404	0.038843744	22.557919	0.000000e+00
alpha1	0.20625680	0.020991560	9.825701	0.000000e+00
alpha2	0.12928037	0.019799283	6.529548	6.596856e-11
alpha3	0.07238938	0.016725025	4.328208	1.503278e-05
alpha4	0.14633323	0.019671749	7.438750	1.016964e-13
alpha5	0.18568246	0.022026744	8.429864	0.000000e+00

6. For both the daily and monthly models, obtain the correlograms of the squared residuals. Also, conduct an ARCH(7) test for each of these models.

Similar to identifying the lag structure of the mean equation, we can iteratively use ARCH tests to determine the lag structure of the ARCH effects, or use a correlogram of the squared residuals. We use squared residuals since we are looking for dynamic patterns in the variance, and the squared residuals approximate the variance.

For leftover ARCH, we use the correlogram of the squared standardised residuals. If there is a pattern, there is leftover ARCH. The correlograms for the squared (standardised) residuals of both series are below:

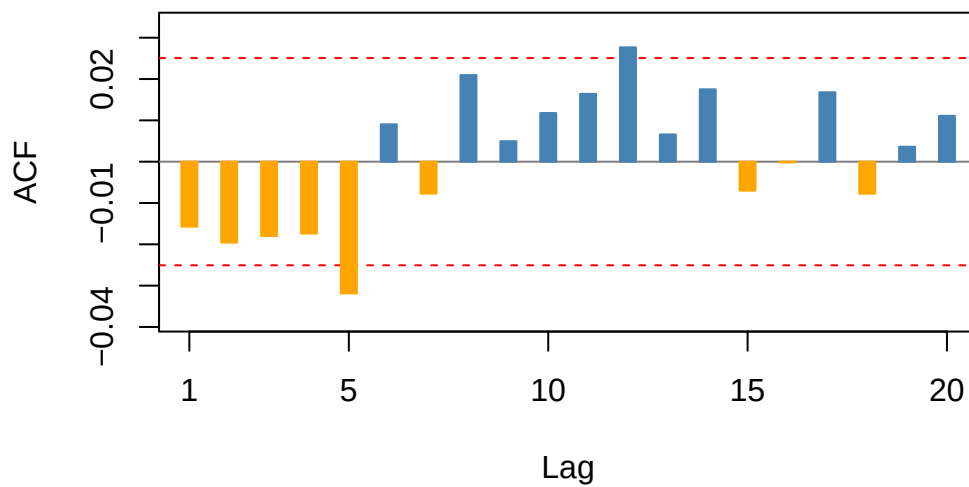
```
# correlogram of standardized squared residuals from ARCH5
res_s.arch5d <- residuals(fit_arma11arch5d, standardize = TRUE)

# square residuals
res_s.arch5d_sq <- res_s.arch5d^2

correlogram(res_s.arch5d_sq,max.lag = 20,
             main.acf = "Correlogram of Daily Squared Standardised Residuals")
```

Lag	AC	Q	Prob
1	-0.0157	1.5110	0.21900
2	-0.0196	3.8567	0.14540
3	-0.0180	5.8317	0.12010
4	-0.0174	7.6819	0.10390
5	-0.0319	13.8988	0.01627
6	0.0090	14.3992	0.02548
7	-0.0077	14.7649	0.03913
8	0.0210	17.4528	0.02572
9	0.0050	17.6034	0.04006
10	0.0118	18.4529	0.04779
11	0.0164	20.1038	0.04394
12	0.0277	24.7915	0.01584
13	0.0066	25.0579	0.02268
14	0.0175	26.9370	0.01962
15	-0.0070	27.2378	0.02686
16	-0.0002	27.2380	0.03891
17	0.0168	28.9681	0.03482
18	-0.0077	29.3353	0.04443
19	0.0036	29.4153	0.05973
20	0.0111	30.1698	0.06715

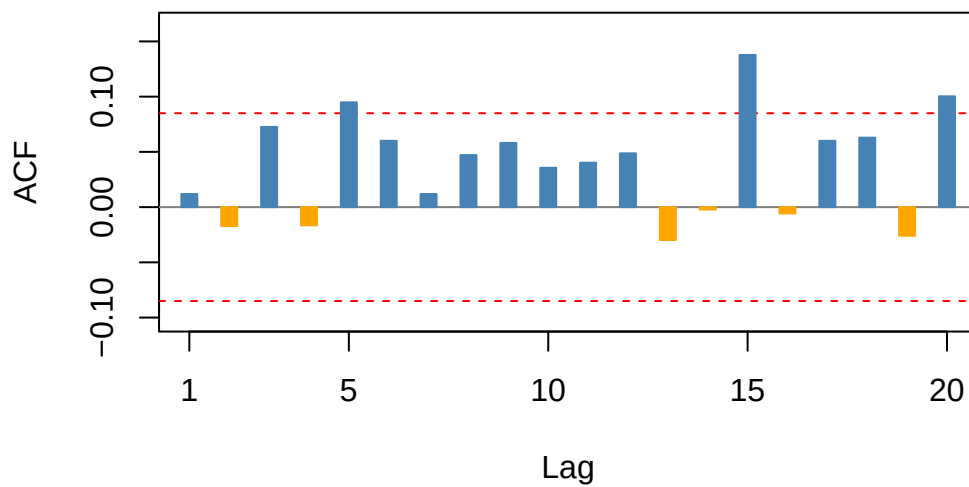
Correlogram of Daily Squared Standardised Residuals



```
# standardised residuals squared from ARCH2
res_s.arch2m_sq <- res_s.arch2m^2
correlogram(res_s.arch2m_sq,max.lag = 20,
             main.acf = "Correlogram of Monthly Squared Standardised Residuals")
```

Lag	AC	Q	Prob
1	0.0119	0.0751	0.78400
2	-0.0172	0.2331	0.89000
3	0.0726	3.0659	0.38160
4	-0.0165	3.2126	0.52290
5	0.0948	8.0619	0.15290
6	0.0600	10.0087	0.12430
7	0.0119	10.0851	0.18380
8	0.0471	11.2887	0.18590
9	0.0581	13.1243	0.15710
10	0.0356	13.8143	0.18160
11	0.0402	14.6961	0.19680
12	0.0487	15.9900	0.19170
13	-0.0298	16.4752	0.22440
14	-0.0022	16.4778	0.28510
15	0.1378	26.9077	0.02950
16	-0.0057	26.9256	0.04232
17	0.0600	28.9134	0.03533
18	0.0629	31.0975	0.02805
19	-0.0259	31.4691	0.03583
20	0.1003	37.0521	0.01153

Correlogram of Monthly Squared Standardised Residuals



What do the correlograms suggest? Is there left-over ARCH?

The left over ARCH(7) test for the daily series is (remember this is using the ARCH(5) residuals):

```
# Left over ARCH(7) test from ARCH 5 model
# standardised residuals

FinTS::ArchTest(res_s.arch5d, lags = 7)
```

ARCH LM-test; Null hypothesis: no ARCH effects

```
data: res_s.arch5d
Chi-squared = 15.762, df = 7, p-value = 0.02738
```

For monthly (using the ARCH(2) residuals):

```
# Left over ARCH(7) test
FinTS::ArchTest(res_s.arch2m, lags = 7)
```

ARCH LM-test; Null hypothesis: no ARCH effects

```
data: res_s.arch2m
Chi-squared = 10.131, df = 7, p-value = 0.1813
```

What do the test results suggest? What should you do next?

7. What does this tell us about the number of required ARCH terms needed to successfully capture conditional heteroskedasticity?

Summarise the results.

8. Why is the same behaviour not observed in the monthly series?

See if your group can work this out